

Post Lab

Consider a set of processes with burst times 10, 5, 8, 6, and 2 units. The processes arrive in the system in the given order. Write a program to calculate the waiting time incurred while servicing these processes using FCFS and SJF scheduling algorithms, and compare the performance of both algorithms.

```
GNU nano 8.7
#include <stdio.h>

int main() {
    int bt[5] = {10, 5, 8, 6, 2};
    int wt_fcfs[5], wt_sjf[5];
    int i, j;
    wt_fcfs[0] = 0;
    for(i=1; i<5; i++)
        wt_fcfs[i] = wt_fcfs[i-1] + bt[i-1];

    printf("FCFS Waiting Times:\n");
    for(i=0; i<5; i++)
        printf("%d ", wt_fcfs[i]);

    // SJF sorting
    for(i=0; i<5; i++) {
        for(j=i+1; j<5; j++) {
            if(bt[i] > bt[j]) {
                int temp = bt[i];
                bt[i] = bt[j];
                bt[j] = temp;
            }
        }
    }

    wt_sjf[0] = 0;
    for(i=1; i<5; i++)
        wt_sjf[i] = wt_sjf[i-1] + bt[i-1];

    printf("\nSJF Waiting Times:\n");
    for(i=0; i<5; i++)
        printf("%d ", wt_sjf[i]);

    return 0;
}
```

^G Help ^O Write Out ^F Where Is ^K Cut
^X Exit ^R Read File ^\ Replace ^U Paste

```
De11@aryaaa MSYS ~  
$ nano compare.c  
  
De11@aryaaa MSYS ~  
$ gcc compare.c -o compare  
  
De11@aryaaa MSYS ~  
$ ./compare  
FCFS waiting Times:  
0 10 15 23 29  
SJF waiting Times:  
0 2 7 13 21  
De11@aryaaa MSYS ~  
$ |
```